TECHNICAL DOCUMENTATION

FORGE

Task Decomposition & Technical Specification

Multi-Agent Architecture • Agent Contracts • Decision Logic • Data Flow

Version	Authors	Architecture	LLM
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1. System Overview & Design Philosophy

FORGE is structured as a four-stage sequential multi-agent pipeline. Each stage is a self-contained agent with a clearly defined input contract, processing responsibility, and output specification. The sequential architecture ensures that each downstream agent operates on the verified output of its predecessor, creating a compounding chain of informed, consistent, and coherent plan generation.

Step	Agent	Tool(s)	Input	Output
01	**Guidelines Researcher**	Gemini + Tavily	User profile (all fields)	Structured research findings: exercise + nutrition guidelines
02	**Workout Planner**	Gemini	Profile + Research output	4-week progressive workout schedule (sets, reps, days, progression)
03	**Nutrition Advisor**	Gemini	Profile + Research output	Macro targets, 7-day meal plan, hydration & supplement guide
04	**LLM-as-Judge**	Gemini	Profile + Workout + Nutrition	Scientific score (0–10), rubric breakdown, strengths, warnings

2. User Profile Input Specification

The user profile is the single source of truth that all four agents consume. It is collected via the Streamlit UI and propagated to each agent as structured context. All fields are used by at least one agent; several fields trigger conditional logic in multiple agents.

Field	Type	Values / Format	Agent(s) Using	Triggers Conditional Logic?
Fitness Goal	Enum	Weight Loss, Muscle Gain, Endurance, General Wellness	All agents	Yes — shapes exercise selection & macro targets
Fitness Level	Enum	Beginner, Intermediate, Advanced	Researcher, Workout Planner, Judge	Yes — determines exercise complexity & volume
Age	Integer	Numeric input (years)	Researcher, Nutrition Advisor	Yes — affects caloric targets & recovery needs
Weight	Float	kg or lb	Nutrition Advisor	Yes — macro calculation baseline
Available Equipment	Multi-select	Dumbbells, Barbell, Resistance Bands, Bodyweight only, etc.	Workout Planner	Yes — bodyweight-only fallback if none selected
Dietary Preference	Enum	None, Vegetarian, Vegan, Keto, Paleo, etc.	Researcher, Nutrition Advisor, Judge	Yes — enforced compliance; 1.5x weight in rubric
Health Notes	Free text	Open field for injuries, conditions, restrictions	Workout Planner, Judge	Yes — movement contraindication avoidance
Training Days/Week	Integer	1–7 days	Workout Planner	Yes — determines weekly schedule structure

3. Agent-Level Specifications

Agent 1 — Guidelines Researcher

Role	Retrieves current, evidence-based fitness and nutrition guidelines relevant to the user's specific profile from the live web.

Tools	Google Gemini 1.5 Flash (reasoning + synthesis) + Tavily Search API (real-time web retrieval)

Trigger	Executes immediately upon form submission with the complete user profile.

Input Contract	Full user profile: goal, fitness level, age, weight, equipment, dietary preference, health notes, training days.

Processing Logic	Formulates targeted Tavily search queries based on user profile dimensions. Retrieves and evaluates search results. Synthesizes findings into structured guidelines covering exercise recommendations and nutritional principles.

Output Contract	Structured text containing: (a) evidence-based exercise guidelines for the user's goal and level, (b) nutritional guidelines including macro principles, caloric guidance, and dietary recommendations, (c) source references for grounding.

Failure Handling	If Tavily returns insufficient results, agent falls back to Gemini's internal knowledge with a note that results are not web-grounded. UI displays a warning label.

Dependencies	Tavily API key (user-provided). Google Gemini API key (user-provided). Internet connectivity.

Agent 2 — Workout Planner

Role	Generates a structured, progressive 4-week workout schedule tailored to the user's equipment, goals, fitness level, health conditions, and training availability.

Tools	Google Gemini 1.5 Flash

Trigger	Executes upon successful completion of Agent 1. Receives Agent 1 output as part of its context.

Input Contract	User profile (all fields) + Agent 1 research findings.

Processing Logic	Applies progressive overload principles across 4 weeks (volume or intensity increase per week). Structures workouts per available training days (e.g., 3-day push/pull/legs, 5-day split). Selects exercises constrained to available equipment. Excludes movements contraindicated by health notes. Specifies sets, reps, rest periods, and weekly progression targets.

Output Contract	4-week workout plan structured by week and day: exercise name, sets × reps, rest period, coaching notes, and weekly progression rationale.

Decision Point A	No equipment selected → Automatically defaults to a bodyweight-only exercise library with no fallback exceptions.

Decision Point B	Health notes present → All exercises are pre-screened against the stated condition. Contraindicated movements are replaced with safe alternatives. Screening logic is made explicit in the output.

Decision Point C	Training days ≤ 2 → Full-body workouts are prioritized over splits to maximize muscle group frequency within the constraint.

Agent 3 — Nutrition Advisor

Role	Generates a complete, personalized nutrition plan including daily macro targets, a 7-day meal plan, hydration guidelines, and supplement recommendations.

Tools	Google Gemini 1.5 Flash

Trigger	Executes upon successful completion of Agent 1. Operates in parallel with Agent 2 logically but receives Agent 1 output as context.

Input Contract	User profile (all fields, with weight and dietary preference as primary drivers) + Agent 1 research findings.

Processing Logic	Calculates estimated Total Daily Energy Expenditure (TDEE) using age, weight, and activity level derived from training days. Applies goal-specific caloric adjustment (deficit for fat loss, surplus for muscle gain). Distributes macros (protein, carbohydrates, fats) per evidence-based guidelines from Agent 1. Constructs a 7-day meal plan with meal timing and food variety. Generates hydration targets and conditionally recommends supplements.

Output Contract	Daily macro targets (calories, protein g, carbs g, fat g), 7-day meal plan (breakfast, lunch, dinner, snacks per day), hydration target (liters/day), supplement recommendations with rationale.

Decision Point A	Dietary Preference = Vegan/Vegetarian → All meal plan items must be fully compliant. Zero non-compliant ingredients permitted. Protein sources restricted to plant-based options with attention to completeness.

Decision Point B	Dietary Preference = Keto/Paleo → Macro distribution adjusted: higher fat ratio, reduced carbohydrate ceiling. Meal plan restructured accordingly.

Decision Point C	Goal = Weight Loss → Caloric target set to 15–20% deficit below TDEE. Protein target elevated to preserve lean mass during deficit.

Decision Point D	Goal = Muscle Gain → Caloric target set to 10–15% surplus above TDEE. Protein target elevated to support hypertrophy. Meal timing recommendations include pre/post workout nutrition windows.

Agent 4 — LLM-as-Judge

Role	Evaluates the complete generated plan (workout + nutrition) against a 10-criterion scientific rubric to produce a weighted quality score and actionable feedback.

Tools	Google Gemini 1.5 Flash

Trigger	Executes only after both Agent 2 and Agent 3 have completed. Receives all outputs plus the original user profile.

Input Contract	User profile + Agent 2 workout plan + Agent 3 nutrition plan.

Independence Guarantee	The Judge agent prompt is explicitly designed to evaluate the output critically, without access to the generation rationale. It is instructed to score objectively, not to validate.

Output Contract	Numeric score per criterion (0–1.0), applied weight per criterion, weighted sub-score, final aggregate score (0–10), qualitative strengths, qualitative warnings, and an overall recommendation statement.

Decision Point A	Final score < 6.0 → Warning banner displayed prominently in the UI. User advised to consult a professional before following the plan.

Decision Point B	Dietary Preference Compliance criterion fails (score = 0) → Plan is flagged as non-compliant. Regardless of overall score, compliance failure is surfaced as a critical warning.

Decision Point C	Safety & Injury Consideration criterion score < 0.5 → Safety warning displayed even if overall score is ≥ 6.

4. LLM-as-Judge Evaluation Rubric

The Judge agent evaluates plans across 10 scientific criteria. Two criteria — Dietary Preference Compliance and Safety & Injury Consideration — are assigned elevated weights (1.5x) because failures in these dimensions carry disproportionate risk of harm to the user. The remaining eight criteria are equally weighted at 1.0x.

Criterion	Wt.	Evaluation Question	Scientific Rationale
Progressive Overload Principle	1.0x	Does the workout progressively increase volume or intensity week over week?	Foundational principle of adaptation-driving training.
Muscle Group Balance	1.0x	Are all major muscle groups addressed without systematic over/under-emphasis?	Prevents postural imbalances and overuse injuries.

Rest & Recovery Adequacy	1.0x	Is sufficient rest built into the weekly schedule? Are muscle groups allowed adequate recovery time?	Recovery is where physiological adaptation occurs.
Macro Distribution Validity	1.0x	Are protein, carbohydrate, and fat targets within evidence-based ranges for the stated goal?	Macros are the primary nutritional driver of body composition.
Evidence-Based Exercise Selection	1.0x	Are chosen exercises supported by exercise science for the stated goal?	Ensures efficacy and relevance of the exercise selection.
Fitness Level Appropriateness	1.0x	Are volume, intensity, and exercise complexity appropriate for the user's stated level?	Prevents injury from inappropriate loading.
Dietary Preference Compliance ⚡	**1.5x**	Does the nutrition plan fully comply with the stated dietary preference without exception?	Non-compliance could cause ethical harm or health risk.
Safety & Injury Consideration ⚡	**1.5x**	Does the plan actively avoid movements contraindicated by stated health conditions?	Non-compliance could cause direct physical harm.
Goal Alignment	1.0x	Is the overall plan coherently oriented toward the user's primary stated goal?	Ensures the plan delivers on its core purpose.
Completeness & Actionability	1.0x	Is the plan specific enough to follow without additional interpretation?	A plan that cannot be acted upon has zero practical value.

⚡ = Elevated weight (1.5x). Failures in these criteria trigger critical safety warnings regardless of overall score.

5. Data Flow & Inter-Agent Communication

User → Agent 1	Complete user profile (all 8 fields as structured context in prompt)
Agent 1 → Agent 2	Research findings (exercise guidelines, goal-specific protocols, evidence citations)
Agent 1 → Agent 3	Research findings (nutritional principles, macro guidelines, dietary research)
Agent 2 → Agent 4	Complete 4-week workout plan (structured text)
Agent 3 → Agent 4	Complete nutrition plan (macro targets + meal plan + recommendations)
Profile → Agent 4	Original user profile (for consistency validation — Judge checks output against stated inputs)

Agent 4 → UI	Score (0–10), rubric breakdown, warnings, recommendation string

6. Technology Stack & Infrastructure

Layer	Technology	Version / Detail	Role in System
Frontend	Streamlit	1.32+	Web UI, form collection, output rendering, warning display
LLM Engine	Google Gemini	gemini-1.5-flash	Reasoning, synthesis, and judgment across all 4 agents
Web Search	Tavily Search API	Latest	Real-time evidence retrieval for Agent 1 (Guidelines Researcher)
Language	Python	3.10+	Core application logic, agent orchestration, API calls
Deployment	Railway	Latest	Cloud hosting, CI/CD pipeline, production environment
Secrets	Streamlit Sidebar	User-provided	Gemini API key + Tavily API key entered at session start

7. Project Structure

File / Directory	Responsibility
`app.py`	Main Streamlit application — UI layout, user input collection, agent orchestration, output rendering, and warning logic.
`agents/guidelines_researcher.py`	Agent 1 — Constructs Tavily queries, retrieves web results, and synthesizes evidence-based fitness & nutrition guidelines via Gemini.
`agents/workout_planner.py`	Agent 2 — Generates the 4-week progressive workout plan with equipment constraints, health note screening, and level-appropriate volume.
`agents/nutrition_advisor.py`	Agent 3 — Calculates TDEE-based macro targets and constructs the full 7-day meal plan with dietary compliance enforcement.
`agents/llm_judge.py`	Agent 4 — Runs the weighted 10-criterion scientific evaluation rubric and produces the final score, breakdown, and recommendation.
`requirements.txt`	Python dependency manifest for deployment.
`Procfile`	Railway deployment configuration specifying the Streamlit run command.